

Pattern Recognition Letters

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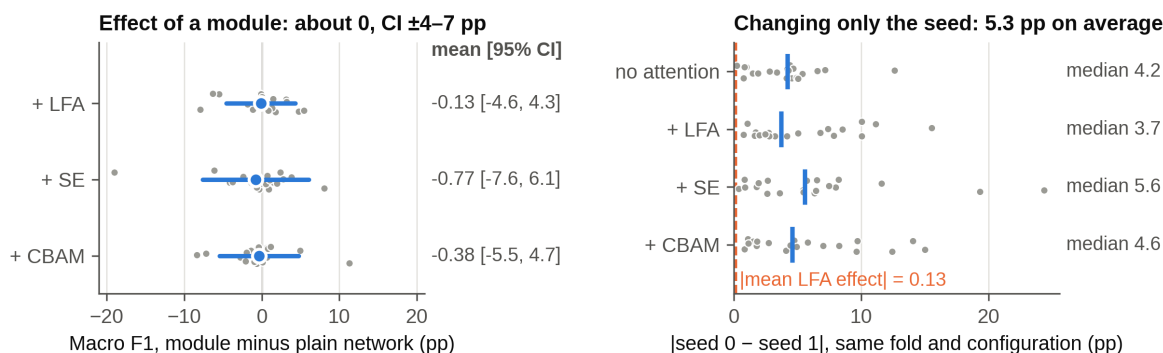
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Seed-to-seed variation outweighs lightweight attention gains: a grouped, repeated cross-validation study on field durian disease images

Lin Ding Shan

Lightweight attention on 550 field durian disease images: 20 grouped folds $\times$ 2 seeds, corrected paired tests

EfficientNet-B0 with LFA (1,281 parameters), SE or CBAM against the plain network. No module met the pre-registered decision rule; the seed alone moved macro F1 by more than the 2-pp threshold. Single-run comparisons of a few points on data of this size are not informative.



EfficientNet-B0 was trained on 550 field images of four durian disease classes with and without three lightweight attention modules (LFA, SE, CBAM) on 20 capture-grouped folds with two seeds each, and compared in pairs with the corrected resampled t-test. No module met a decision rule fixed before the runs (LFA: -0.13 percentage points, 95% CI -4.56 to 4.31), while changing only the random seed moved macro F1 by 5.3 points on average. Single-run gains of a few points on data of this size are not evidence about the architecture.

Research Highlights (Required)

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- Three attention modules tested on 550 field durian images: 20 grouped folds, 2 seeds
- No module met a pre-registered rule; LFA vs plain net -0.13 pp (95% CI -4.56 to 4.31)
- Changing only the random seed moved macro F1 by 5.3 points on average
- Earlier insertion points and 320-px input did not help in a 200-run follow-up
- Single-run gains of a few points on small field datasets are not evidence